Small-prime transfers and an algebraic shortcut

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Standing: unsigned producer assertions. The opt-in small-prime backend supports checked odd primes≤4096, with exhaustive enumeration capped at101. It imports the unchanged repository Fp2 context; this is explicit arithmetic reuse, not independent validation of that layer. There is no benchmark or cryptographic-size field claim. [pcc:unverified]

Fixed correctness controls

p	Non-residue	Each Jac order	Rational 2-torsion	Source E order	r; generator
13	2	180	4 / 4	10	5; (10,6)
29	2	960	4 / 16	40	5; (24,15)
31	3	1008	16 / 4	28	7; (19,2)
101	2	9216	4 / 16	96	3; (48,20)

The original13/29/101 fields were fixed before execution. The separate31 control varies the automatic nonresidue and tests a descending2-torsion case. Every rational class, both dual norms, equal fibers and odd-subgroup scalar relations pass. Both F101 hops agree exhaustively with the previous model backend. Equal2-torsion counts at13 do not establish an isomorphism. [pcc:unverified]

The final feature suite passes25 tests and strict Clippy; the root default example import check passes. The unchanged BigUint reference compares all eight complete class sets, independently counts Jacobian orders and self-inverse elements, and checks additions and dual norms. Its chart adapter uses BigUint directly. Independent elliptic quotient orders are10/18,40/24, 28/36,96/96. The Fp2 counter shares the existing algorithm lineage, so no signed independence is inferred. [pcc:unverified]

Receipts are frozen in evidence/, with runtime inputs in small-prime-inputs.sha256. Claimscd7c2f and cryptanalysisdb1613 are unverified. The first family reference run is development evidence; an immutable source pin/replay remains a separate step. Earlier reports and raw failures remain retained with their original scope.

Algebraic endpoint control

For the fixed c=4 source, in every characteristic except2,3,5: $C_s: y^2=(x^2-1)(x^2+x+4)(x^2-x+4)$, and $C_t: w^2=(z^6-102z^4+393z^2-4)/500$. Normalize the target quotient by $X=z^2/500, V=w/500$. Shift source $U=u-1$; its standard2-isogeny is $\phi(U,v)=(U+9+24/U, v(1-24/U^2))$. The codomain isomorphism is $\tau(X,V)=(X/100+1/125, -V/1000)$. The written argument gives $n R f^* = \tau \phi [2]$. It includes the actual base-image translation by (1/125,0). [pcc:unverified]

Seven cleared ring lemmas compile with the standard axiom gate. Total proof inventory is25 written lemmas,22+8+7 audited declarations. Full projective and Picard geometry is a written argument using the primary sources, not a Lean construction. Claims ec195a and6f6202 remain unsigned. This result concerns the first elliptic projection; it is not asserted as a collapse of the later Jacobian-valued two-hop map. [pcc:unverified]

The fiber calculation and remaining work

On the generic correspondence fiber put $a=x^2-1, b=x^2+x+4, c=x^2-x+4$, $A=x^2-x-6, D=9x^2+x-4, M=2x^3+19x^2+7x+74$, $N=8x^3+x^2-47x-4$. The fiber equation is $A\,z^2+10b\,z-D=0$. Its projected two elliptic points lie on $V=mX+n$ with $m=aM/(10yA), n=-aN/(5000yA)$.

The Lean-checked identities simplify their sum to $X_0=1/125-3a/(20bc), V_0=-3y(a^2-24)/(200b^2c^2)$. The source base's correspondence image projects to $T=(1/125,0)$, so it must be subtracted. After translation by T the coordinates become $X=1/125+bc/(100a), V=-y(a^2-24)/(1000a^2)$, exactly tau phi(f(P)). Agreement on this dense open set extends along the actual morphisms of smooth projective curves. Linearity on complete degree-two fibers then gives $n\,R\,f^* = \text{tau phi}$ [2]. The extension/Picard step is explained in GLOBAL_SUCCESSOR_COLLAPSE.md and uses the cited primary geometry; it is not established by the ring lemmas alone. [pcc:unverified]

Native checks cover every rational affine curve point, both original infinities, the actual base image and all elliptic whole fibers over 13,29,31,101. The second target norm is zero on these whole-fiber images. These controls retain degree drops and ramification; finite agreement is separate from the written geometric argument. [pcc:unverified]

Reproduction

Use native Cargo feature small-prime, strict Clippy and the repository proof gate on ChainTransfer.lean,PolynomialIdentities.lean,SuccessorCollapse.lean. The native package remains dependency-free but imports the external unchanged src/prime_hyperelliptic/fp2.rs file. CI checkouts and consumer source guards must include that source. The focused crypto workflow is not yet registered on main; do not claim a feature-branch dispatch ran there.

Cryptanalysis PR1469/1471 are merged. PR1473 freezes the prior two-hop checkpoint at aabae197, with28 local tests and1 existing ignored benchmark. The small-prime feature requires a later explicit dependency/source pin. No field-size extrapolation follows from the four bounded fixtures.

Open gates and next tasks

S3 checkpoint admission credentials and live remaining IsoLab budget/capacity are unavailable. No benchmark has run. KB/ecdb access is unavailable and crypto-autoresearcher is vault locked; its handoff remains outside the vault. Crypto's current-main ledger-prefix fork and historical missing benchmark receipts still block merge. Owner rules require extending the existing PR and preserving both histories, so no alternate ledger lineage or rule bypass has been introduced. The long full-library run remains separate from the passing focused checks and has reported untouched historical failures.

Next: immutable feature oracle replay; examine the simple quadratic cover's restriction for elliptic full groups of odd order and test an alternative odd-degree cover if warranted. Prime-field descent and computational advantage remain open. The explicit shortcut is a required comparison arm; functional equivalence itself is not a measured speed ratio.

Primary sources: Galbraith corrected book ch.8 §8.3,ch.10 §§10.3–10.4/10.6; Kunzweiler,DCC92(2024),1761–1802,DOI10.1007/s10623-024-01366-1, Propositions7–8. No novelty claim is made.